Recap Checkpoint 3 — after 1.4 (Data Types, Data Structures & Boolean Algebra) — cumulative 1.1–1.4 + 2.1

Name: ______________________ Date: __________ Mark: ______ / 50

OCR H446 cumulative recap — mixes every topic taught so far. Show all working.

Questions

Q1. State **one** difference between **CISC** and **RISC** processors, referring to the instruction set. **[2]** — *1.1.2*

Q2. Explain what is meant by **virtual memory** and give **one** drawback of relying on it heavily. **[3]** *(AO1/AO2) — 1.2.1*

Q3. Place the four stages of compilation in the correct order, and name the stage that **removes redundant instructions to make the object code more efficient**: *code generation, lexical analysis, optimisation, syntax analysis*. **[3]** — *1.2.2*

Q4. A company hashes stored passwords but **encrypts** the customer data it transmits. Explain, referring to the **key difference between hashing and encryption**, why each technique is appropriate for its purpose. **[3]** — *1.3.1*

Q5. Convert the denary number **89** into: (a) 8-bit binary **[1]**

(b) hexadecimal **[2]**

Show your working. **[3 total]** — *1.4.1*

Q6. Using **8-bit two's complement**, calculate **39 − 52**. You must: (a) show 39 and 52 in 8-bit binary **[1]**

(b) form the two's complement of 52 **[1]**

(c) perform the addition and state the final **denary** result **[2]**

[4 total] — *1.4.1*

Q7. A floating point value is shown as `0.0001101 × 2⁴` (mantissa with the point after the sign bit, exponent in denary). (a) Normalise the mantissa and state the new exponent. **[2]**

(b) Explain **why** numbers are normalised. **[1]**

[3 total] — *1.4.1*

Q8. Simplify the Boolean expression **$A \wedge (\neg A \vee B)$** using Boolean identities. Show each step and name the identities used. **[3]** — *1.4.3*

Q9. Complete the truth table for **$F = (A \vee B) \wedge \neg(A \wedge B)$** and state which single logic gate it is equivalent to. **[4]** — *1.4.3*

A	B	F
0	0	
0	1	
1	0	
1	1	

Q10. A **stack** (implemented in an array) is initially empty. The following operations are applied in order: `push 8, push 3, push 5, pop, push 9, pop`. (a) State the value returned by **each** `pop` operation. **[2]**

(b) State the contents of the stack (bottom → top) after all operations, and explain why a stack is described as **LIFO**. **[2]**

[4 total] — *1.4.2*

Q11. The values **60, 40, 80, 30, 50, 70** are inserted, in that order, into an initially empty **binary search tree**. (a) Draw or describe the resulting tree. **[2]**

(b) State the order of values produced by an **in-order** traversal and what this demonstrates about a BST. **[2]**

[4 total] — *1.4.2*

Q12. An embedded controller for a heart-rate monitor must store readings, process them in real time, and use as little power as possible. The design team must justify, **synoptically**, three decisions: the **processor architecture (RISC vs CISC)**, an appropriate **data structure** to buffer the most recent readings, and whether to use a **lossy or lossless** compression scheme before transmitting stored readings to a doctor. Evaluate suitable choices for **all three**, justifying each against the device's requirements. **[6]** — *synoptic 1.1 + 1.4 + 1.3*

Section D — Computational thinking (2.1)

Q13. An airport is developing software to track passengers' bags from check-in to the aircraft hold. The system must print tag labels, route bags along conveyor belts, scan bags at checkpoints and report any missing bags. (a) Explain how **decomposition** would help the team develop this system, identifying **two** sub-problems from the scenario. **[2]**

(b) The software models each bag as just `tagID, flightNumber, currentLocation`. Explain how this model is an example of **abstraction**, identifying **one** detail of a real bag that it ignores. **[2]**

[4 total] — *2.1.1 / 2.1.3*

Q14. At a ferry terminal, a vehicle is allowed to board only when the boarding lane is open **and** at least one deck space is left. The program uses the Boolean variable `laneOpen` and the integer variable `deckSpaces`. (a) *Thinking logically:* write the **Boolean condition** the program should test before raising the barrier, using the two variables. **[2]**

(b) The terminal opens **three check-in booths that process vehicles concurrently**. State **one benefit** of this concurrent approach and **one problem** that could arise because the booths share the `deckSpaces` variable. **[2]**

[4 total] — *2.1.4 / 2.1.5*